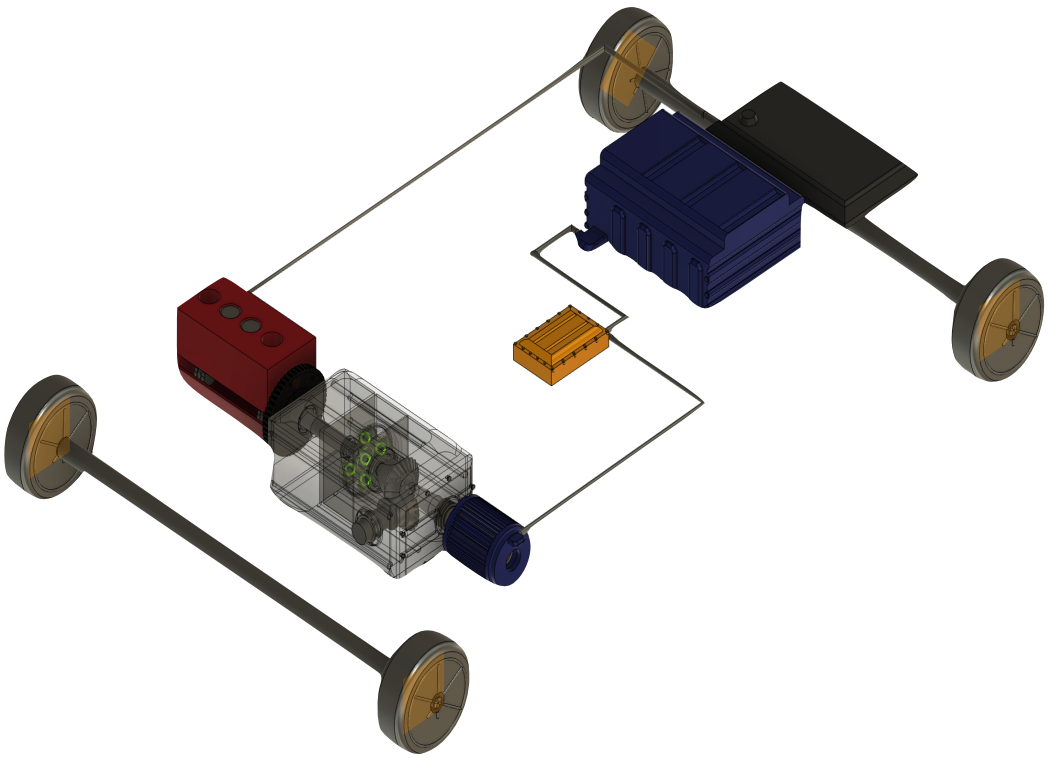
ENGINEERING PORTFOLIO

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Project 01

Hybrid Vehicle Energy and Power Transmission System Modeling

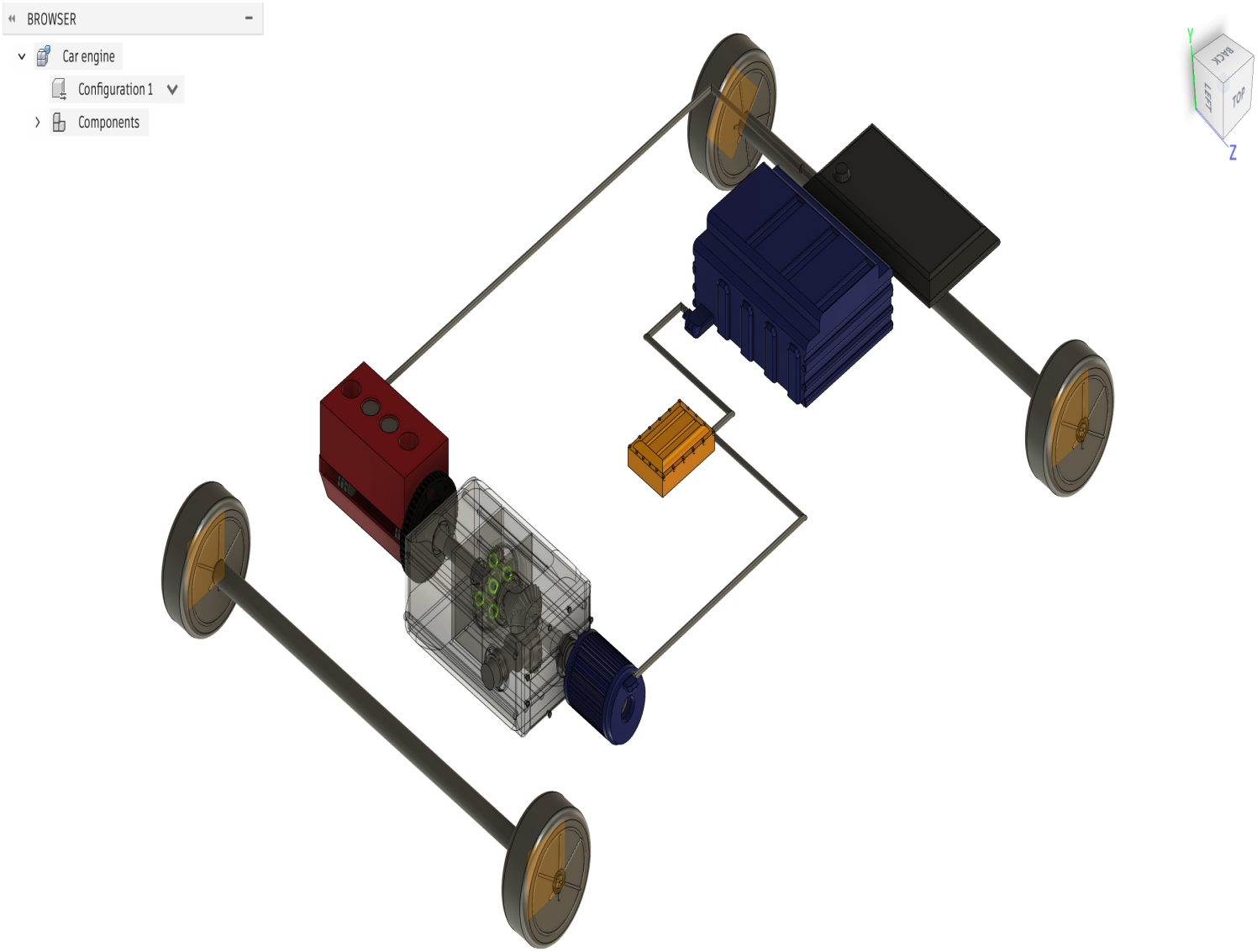
Project Overview

This engineering project focused on the development of a simplified hybrid vehicle model to analyze energy flow and power distribution under different driving conditions. The project combined numerical simulation in MATLAB with 3D mechanical design in Fusion 360 to evaluate hybrid powertrain performance and visualize the overall system architecture.

Objectives

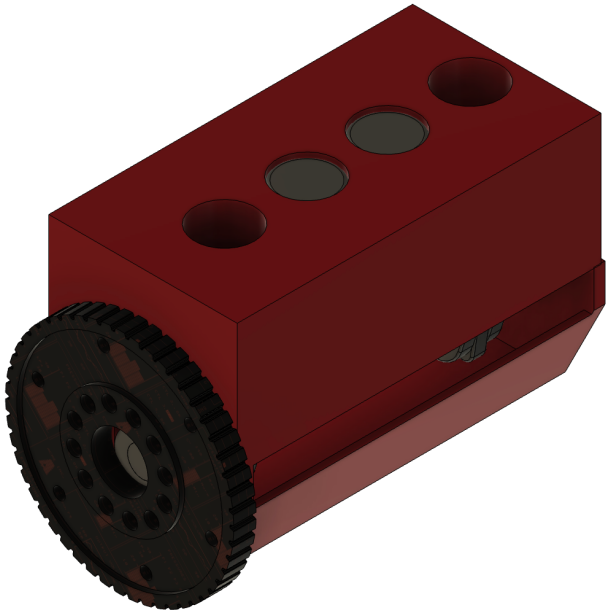
- Develop a simplified hybrid vehicle powertrain model.
- Simulate power distribution between the internal combustion engine and electric motor.
- Analyze battery State of Charge (SOC), fuel consumption, and driving range.
- Design a hybrid powertrain assembly using Fusion 360.
- Evaluate hybrid vehicle performance through engineering simulations.

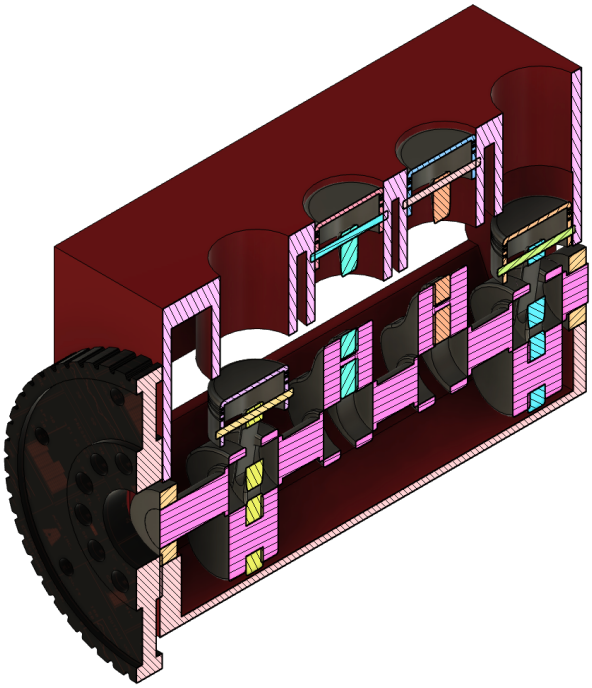
Hybrid Vehicle Assembly



Final hybrid powertrain assembly designed in Fusion 360. The model includes the internal combustion engine, electric motor, transmission, battery, and supporting components used in the MATLAB simulation.

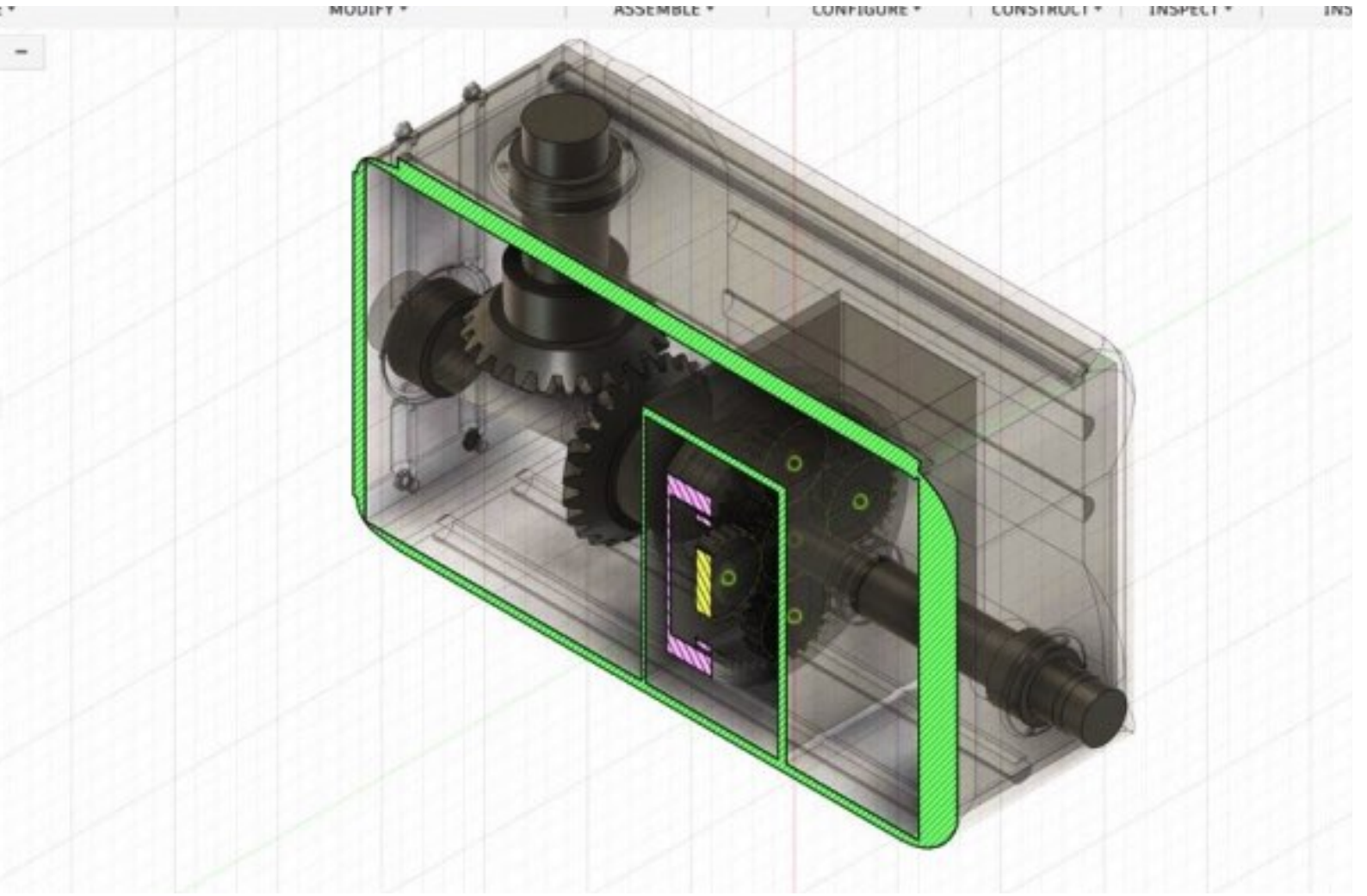
Engine Design





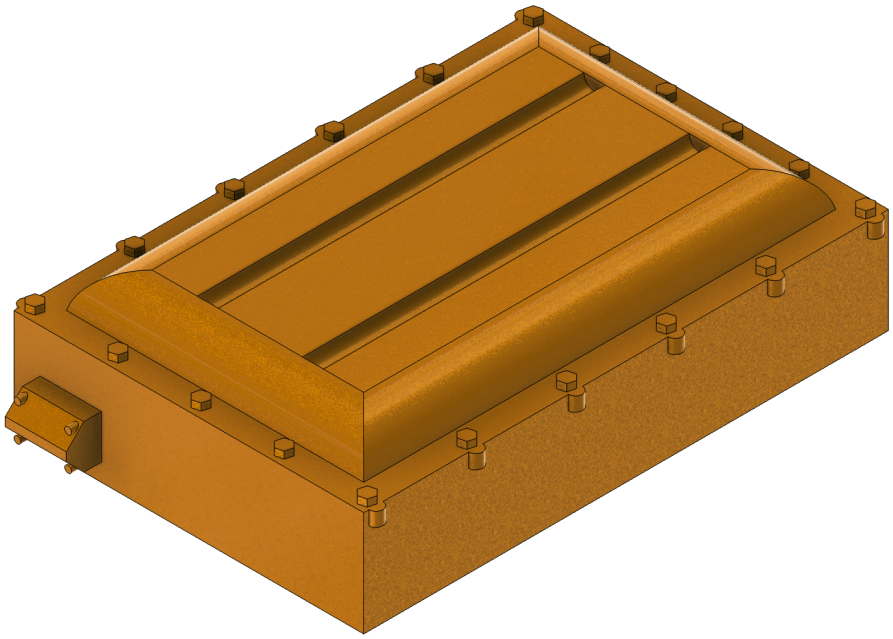
Inline four-cylinder engine modeled in Fusion 360 and integrated into the hybrid powertrain assembly.

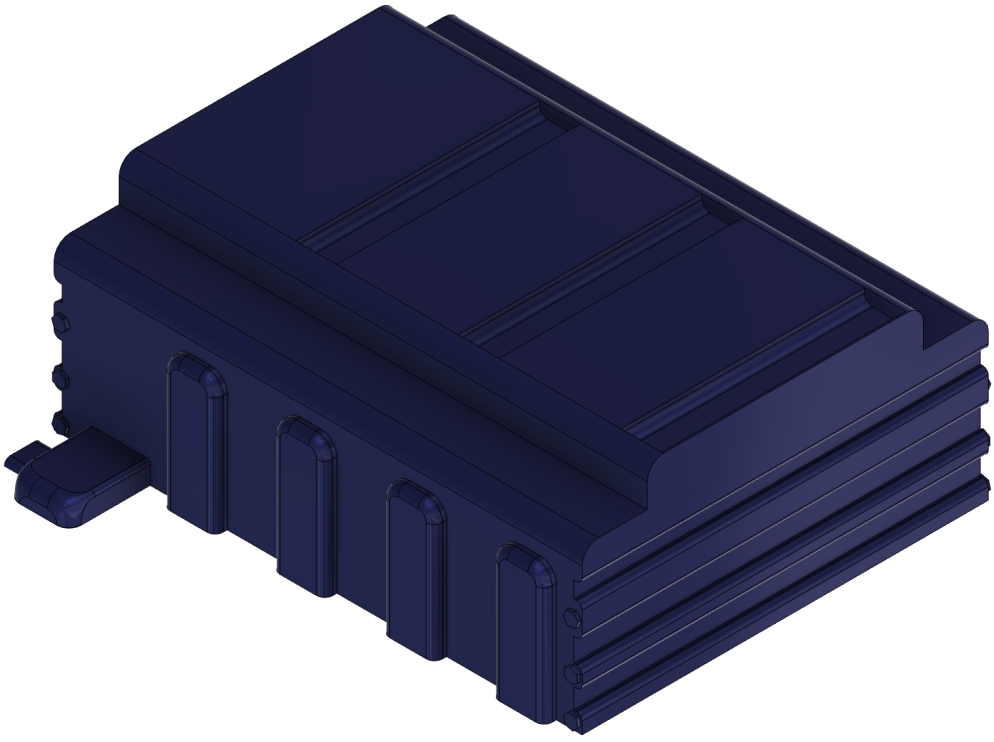
Transmission & Planetary Gear



Planetary gear transmission designed in Fusion 360 to represent the hybrid powertrain's power-splitting mechanism, enabling efficient torque transfer between the internal combustion engine, electric motor, and drivetrain.

Inverter & Battery& Electric Motor







Electrical subsystem of the simplified hybrid powertrain, consisting of the high-voltage battery, inverter, and electric motor. Together, these components provide energy storage, power conversion, electric propulsion, and regenerative braking capability.

MATLAB Simulation

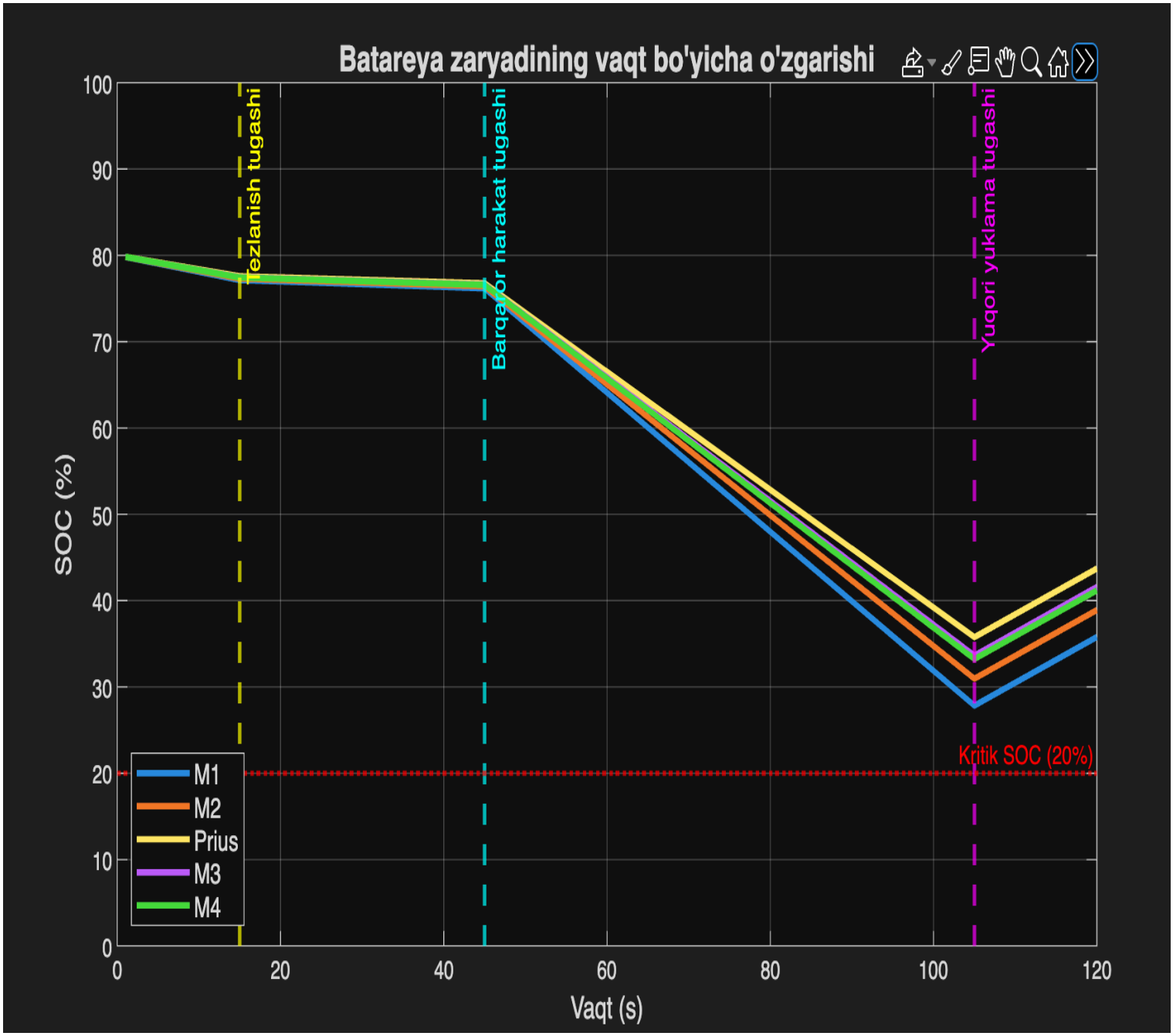
The MATLAB simulation was developed to evaluate the performance of several simplified hybrid vehicle configurations. Four modified models (M1–M4) were created and compared with a baseline vehicle model to investigate the effects of different powertrain configurations on energy management and overall vehicle performance.Also I set up different mode acceleration, cruise, high load and regenerative braking.During the project the mass of the overall car didnt change and the battery power remained the same.

Model	IYOD(kW)	EM(kW)
M1	50	50
M2	62	55
Prius	72	53
M3	75	60
M4	80	65

Simulation Objectives

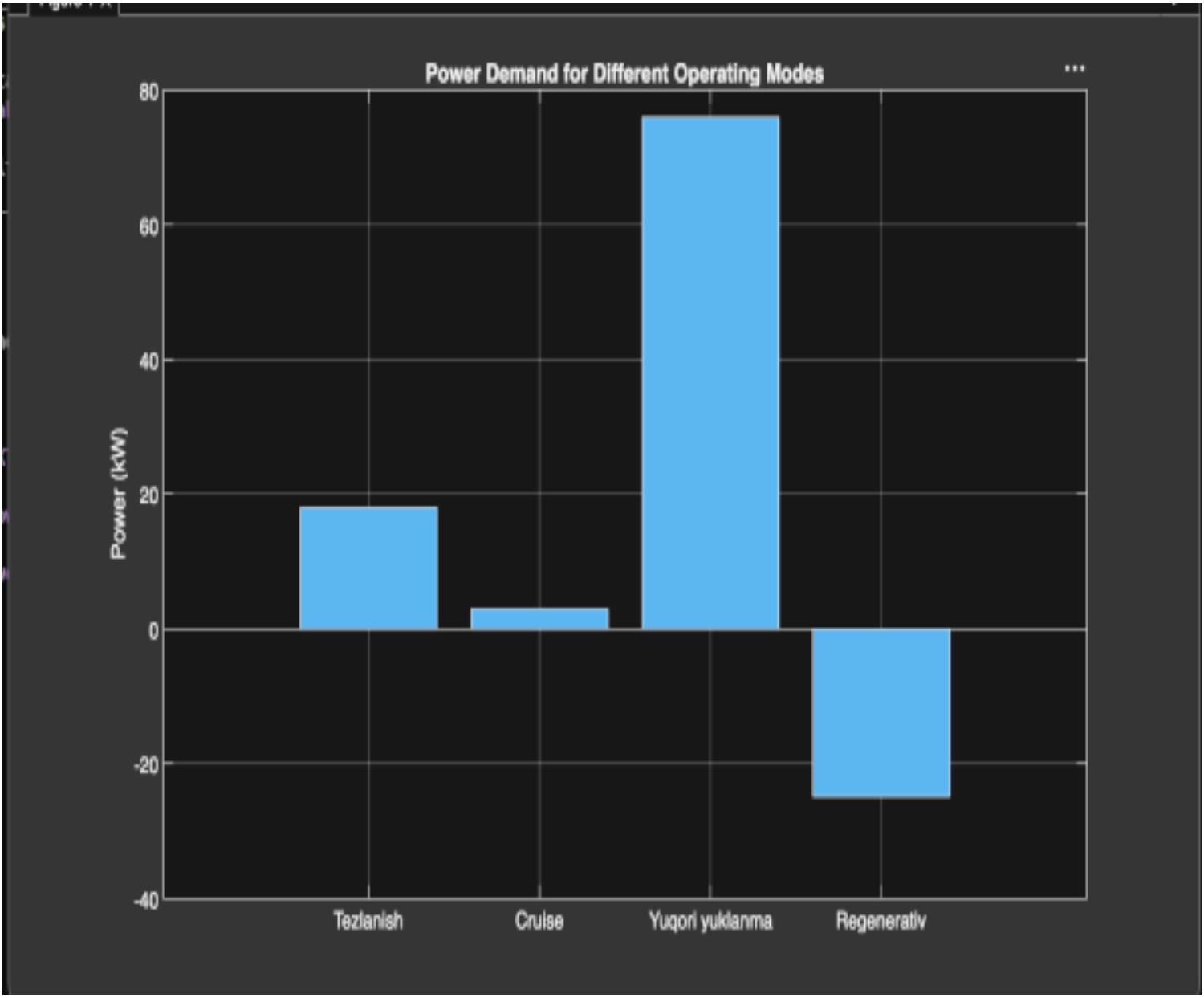
* Develop simplified hybrid vehicle models.
* Compare multiple powertrain configurations (M1–M4).
* Analyze power distribution.
* Evaluate battery State of Charge (SOC).
* Estimate fuel consumption.
* Calculate driving range.

Battery State of Charge graph



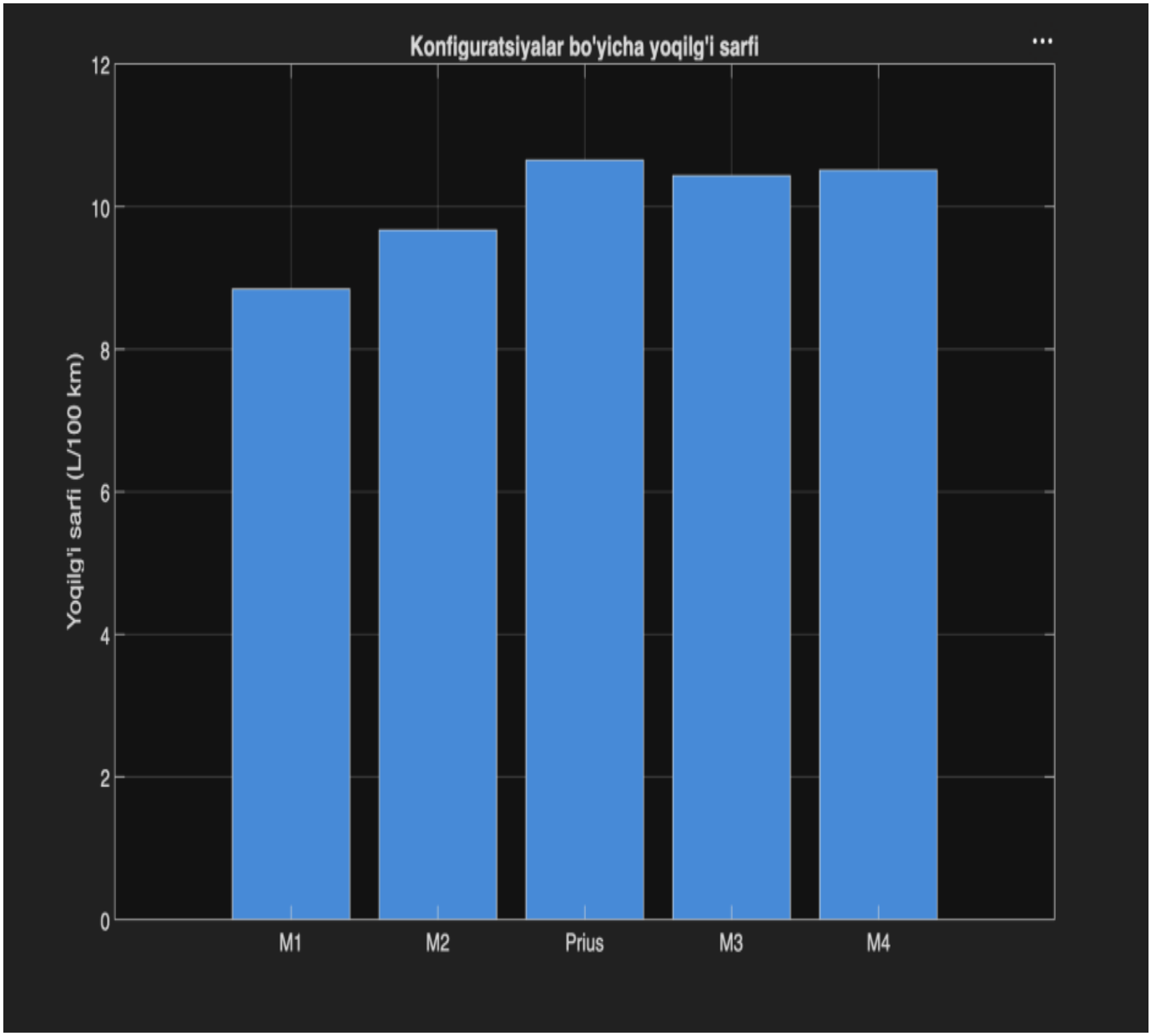
Battery State of Charge (SOC) during the simulation. SOC gradually decreases during vehicle operation and partially recovers through regenerative braking.

Power Graph



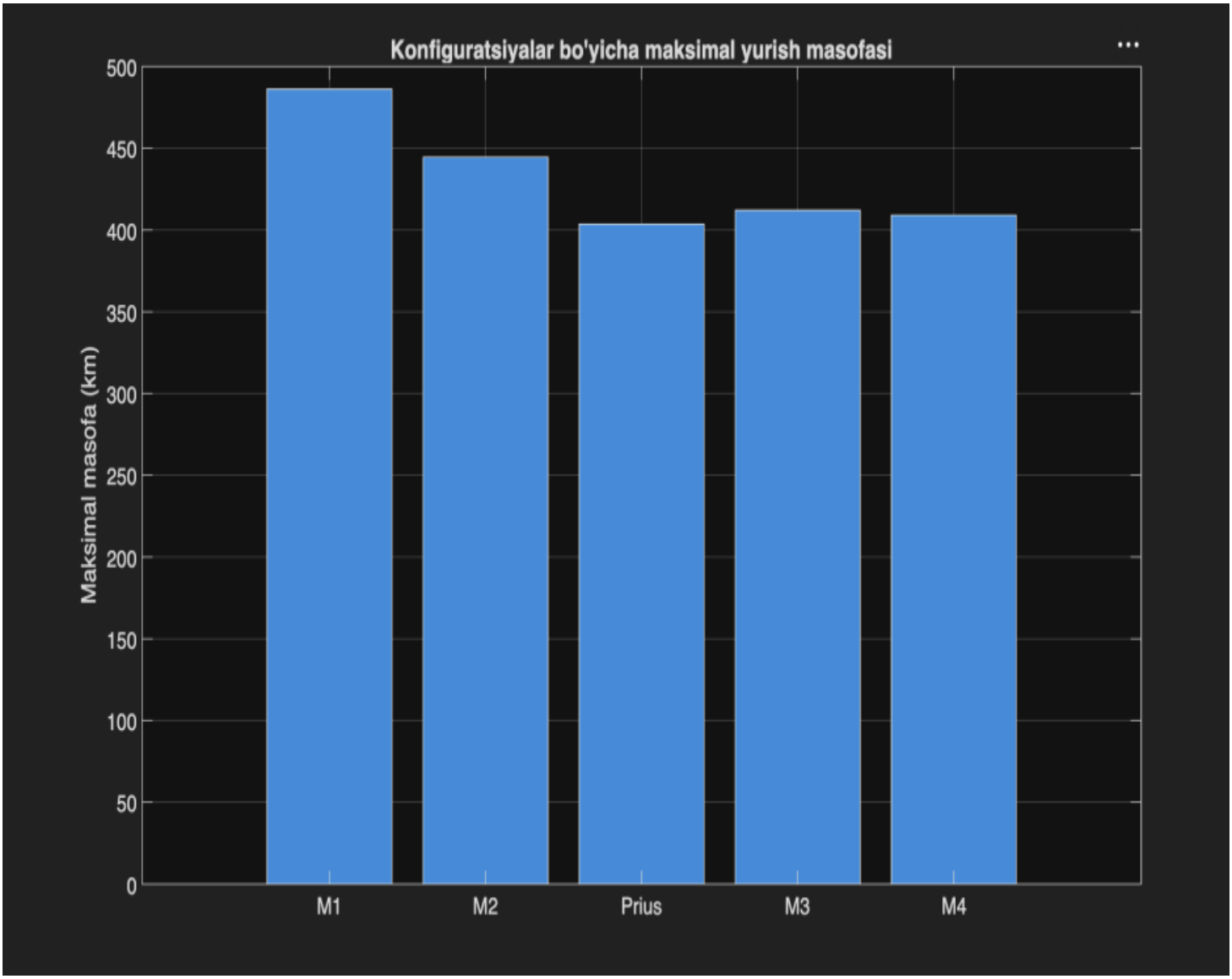
Power distribution between the internal combustion engine and electric motor throughout the simulation under varying driving conditions such as acceleration, cruise, high load and regenerative braking.

Fuel Consumption Analysis



The hybrid powertrain reduced fuel consumption by utilizing the electric motor during low-demand conditions and regenerative braking to recover energy.Because of high load time roughly more than half our estimated time the consumption goes a little higher than we expected.

Driving Range



Estimated driving range based on battery energy usage and fuel consumption throughout the simulation.

Conclusion

The MATLAB simulation successfully demonstrated the operation of a simplified hybrid vehicle powertrain. The results showed effective power distribution between the internal combustion engine and electric motor, realistic battery State of Charge (SOC) behavior, reduced fuel consumption, and an estimated driving range under different operating conditions. The project illustrates the application of engineering simulation and CAD design in evaluating hybrid vehicle performance.